

KARTIK JOSHI

AI Engineer | Generative AI · Agentic AI · LLM & RAG Systems

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SUMMARY

AI Engineer with 3+ years of hands-on experience building and deploying Generative AI, Agentic AI, and RAG systems in production. Currently architect and operate an enterprise Agentic RAG assistant (LangGraph, multi-agent tool-calling, OpenAI/Gemini model routing) deployed on AWS with Docker, alongside supporting deep learning and computer vision experience. Skilled across the full stack: document ingestion and vector search, multi-step agent orchestration, cloud deployment, and measurable latency/cost/accuracy optimization.

SKILLS

Programming Languages: Python, SQL.

AI & Machine Learning: Artificial Intelligence, Machine Learning, Deep Learning, Generative AI, Agentic AI, Multi-Agent Systems, RAG (Agentic RAG), CNNs, Transformer Architectures, Prompt Engineering, Computer Vision, NLP, Tool/Function Calling, Multi-Step Agent Workflows, Embeddings, Semantic & Vector Search, Reranking.

Models & APIs: OpenAI APIs, Google Gemini APIs, Hugging Face Models, Sentence Transformers, Ollama.

Frameworks & Libraries: LangChain, LangGraph, CrewAI, LlamaIndex, LangSmith, LangFuse, MCP (Model Context Protocol), PyTorch, TensorFlow, Keras, FastAPI, Pydantic, OpenCV, NumPy.

Cloud & DevOps: AWS (EC2, ECS, S3, Lambda, Bedrock), GCP (Vertex AI, Compute Engine, Cloud Storage, Document AI), Docker, REST APIs, API Integration, Git & GitHub.

Databases & Infrastructure: ChromaDB, FAISS, Pinecone, MongoDB, Vector Store Infrastructure.

PROFESSIONAL EXPERIENCE

AI Engineer — Easy Home Finance Limited, Mumbai

June 2025 – Present

- Engineered EasyGPT, a document-aware Agentic RAG knowledge assistant adopted by 40+ users across 3 teams — indexing 8K–10K document chunks and handling 1,000+ queries/day, cutting information lookup time by ~60–70%.
- Architected the agentic layer (3 specialized agents, 6 MCP-integrated tools) using LangGraph for multi-step tool-calling orchestration with OpenAI/Gemini model routing, instrumented with LangFuse for tracing and evaluation — reducing latency ~15–20% and inference cost ~20–25% at ~85–90% task success rate.
- Built the core RAG pipeline — document parsing, Sentence-Transformer embeddings, ChromaDB/FAISS retrieval with reranking — and deployed the stack on AWS (EC2, ECS, S3, Lambda) with Docker for production use.
- Developed an Image Forgery Detection pipeline (transformer-based RGB + noise-fingerprint fusion) evaluated on 5,000+ documents, achieving ~91–93% accuracy at ~1–2 sec inference per document.

AI Engineer Intern — Convergence IT Services, Mumbai

Jan 2025 – May 2025

- Built an LLM-powered HRMS chatbot (LangChain, FastAPI, RAG with MongoDB/Pinecone vector store) serving 300+ employees, cutting repetitive HR queries by ~25–30%.
- Built a real-time face recognition attendance system (FaceNet, DeepFace, OpenCV, CNN) for 150+ employees, achieving ~94–96% recognition accuracy and reducing attendance processing time ~40–50%.

AI Engineer Intern — Sharv Auto Tech Pvt Ltd, Mumbai

May 2024 – Dec 2024

- Built an agentic AI chatbot in Python using Generative AI and LLMs to handle conversational, task-oriented interactions.
- Applied NLP techniques (intent recognition, prompt engineering) and explored early AI Agent / tool-calling workflows for AI application development.

AI Engineer Intern — Fafadia Tech, Mumbai

May 2023 – Apr 2024

- Built and fine-tuned CNN and transformer-based deep learning models (PyTorch, TensorFlow, Keras) for image classification, feature extraction, and similarity/transfer-learning tasks.
- Used Docker to containerize model experimentation and deployment workflows, supporting reproducible, version-controlled training pipelines.

PROJECTS

Versify — Agentic AI Content Assistant

- Designed a LangGraph-orchestrated agentic workflow (LangChain, LlamaIndex, LangSmith, CrewAI) routing between text- and image-generation tools; benchmarked 4 LLMs across 300+ prompts (~2–4 sec avg. latency, ~8–9/10 quality score).

Diabetic Retinopathy Detection — Deep Learning

- Built an EfficientNet-B3 classifier on 3,662 APTOS 2019 images (CLAHE, Gaussian filtering, focal loss), reaching ~87–90% accuracy and ~90–93% ROC-AUC; served via FastAPI.

PUBLICATIONS & CERTIFICATIONS

IEEE Conference Publication (Co-Author) — "Retroflex: Uncovering Visual Equivalences through Reverse Image Recon," 11th INDIACom 2024, New Delhi, India | IEEE Xplore

Generative AI Essentials: Using LLMs to Work with Data — IBM SkillsBuild (Sep 2026)

EDUCATION

B.E. in Artificial Intelligence and Data Science — Vidyavardhini College of Engineering and Technology, Mumbai | CGPA: 8.29 | June 2020 – June 2024